

tripleten



Automation of Weather-Based Sales and Staffing Decisions

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Weather- Based Sales and Staffing Decisions

Company: Bayside Brews & Scoops - a coffee and ice cream kiosk located by the beach in Miami, Florida.

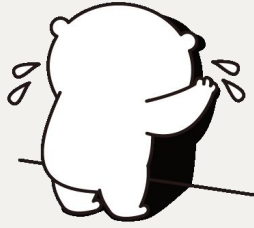
Current challenge: Due to the tropical weather, the company faces business fluctuation that require quick operational adjustments: hot and humid mornings drive surges in iced coffee and gelato sales and windy days near the coast mean fewer beachgoers and more demand for comfort food. Weather changes can happen quickly in coastal cities like Miami, it's best to be prepared.

Manual Process Problems: As it stands, the kiosk manager manually checkers the weather forecasts each morning and then makes decisions about: product mix recommendations (hot drinks vs. cold drinks vs. gelatos vs. pastries), staffing levels based on expected conditions, and customer promotions tailored to the weather.

The manual approach creates several inefficiencies.



Inefficiencies in Manual Approach



1. Time Waste: Manager spends valuable time checking forecasts instead of focusing on operations.
2. Delayed preparations: Manual checking delays important business decisions.
3. Inconsistency: Human oversight can lead to missed opportunities or poor preparation.



Business Impact

Without timely weather-based adjustments, the kiosk risks:

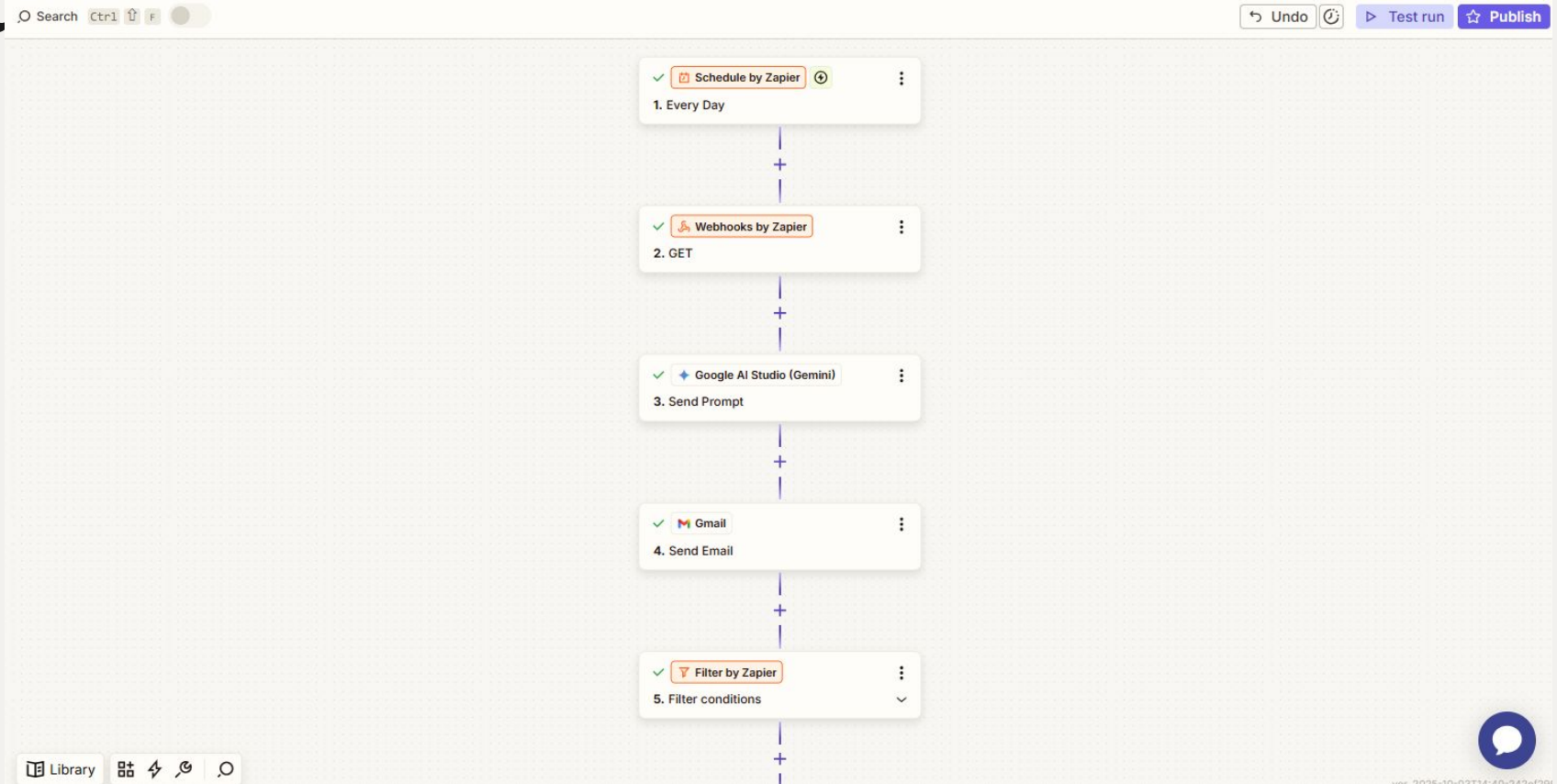
- Lost sales opportunities during peak weather conditions.
- Overstaffing or understaffing based on weather patterns.
- Missing chances to promote weather-appropriate products.
- Potential safety issues during severe weather (temperatures > 91, storms, heavy rain)

Goal

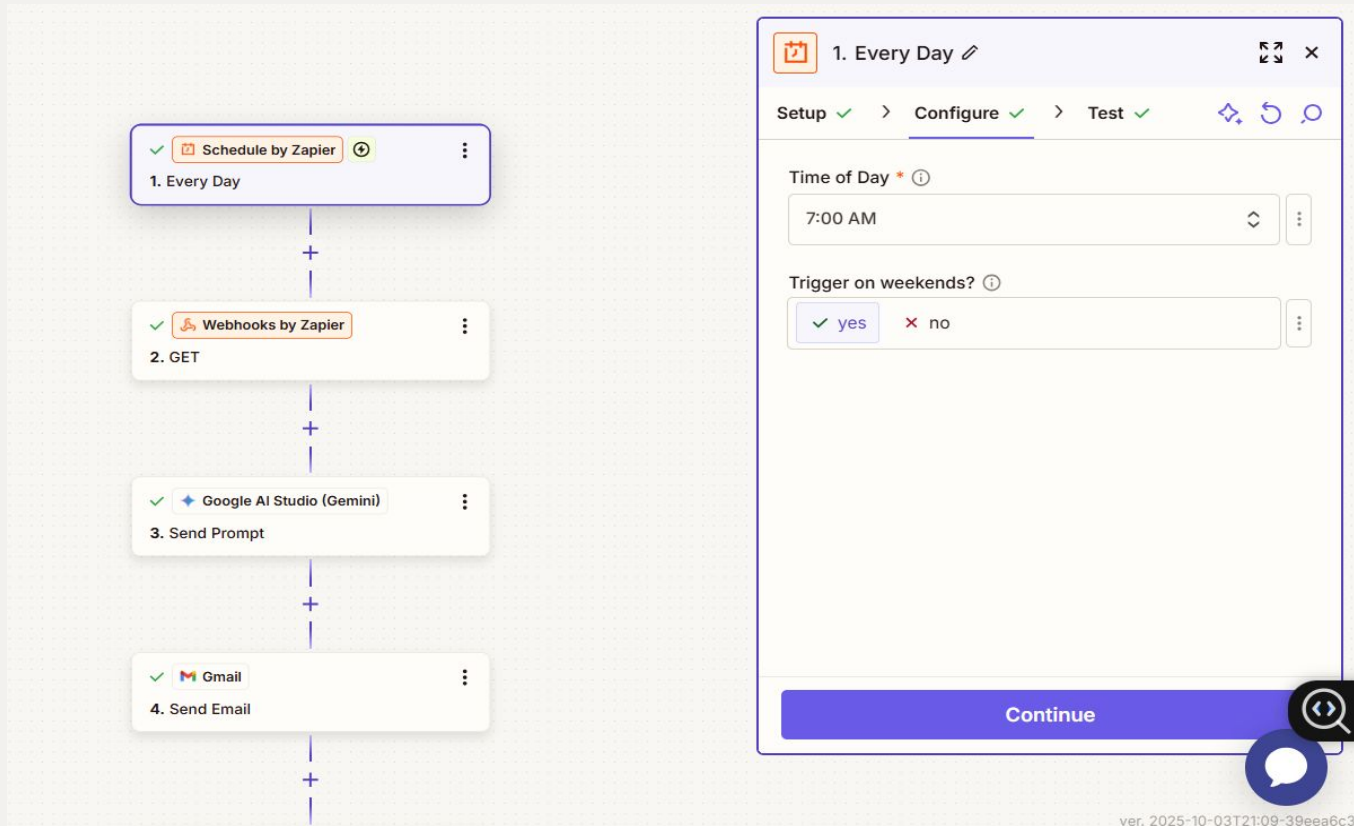
The goal is to automated this entire process so the manager receives actionable insights every morning at 7:00 AM.



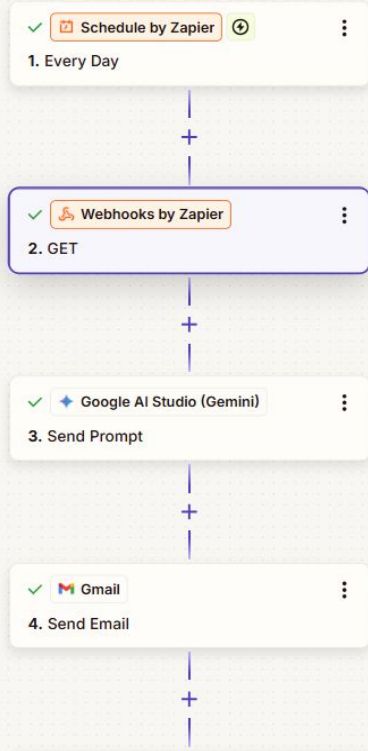
Workflow diagram- Miami Kiosk Weather Guardian



Workflow in action



Workflow in action (cont.)



2. GET

Setup ✓ > Configure ✓ > Test ✓

URL * ⓘ
https://api.openweathermap.org/data/2.5/weather

Query String Params ⓘ

q	Miami, FL, US	+	×
units	Imperial	+	×
appid	69240c6411ce713b40f1ddb4d1523d5	+	×

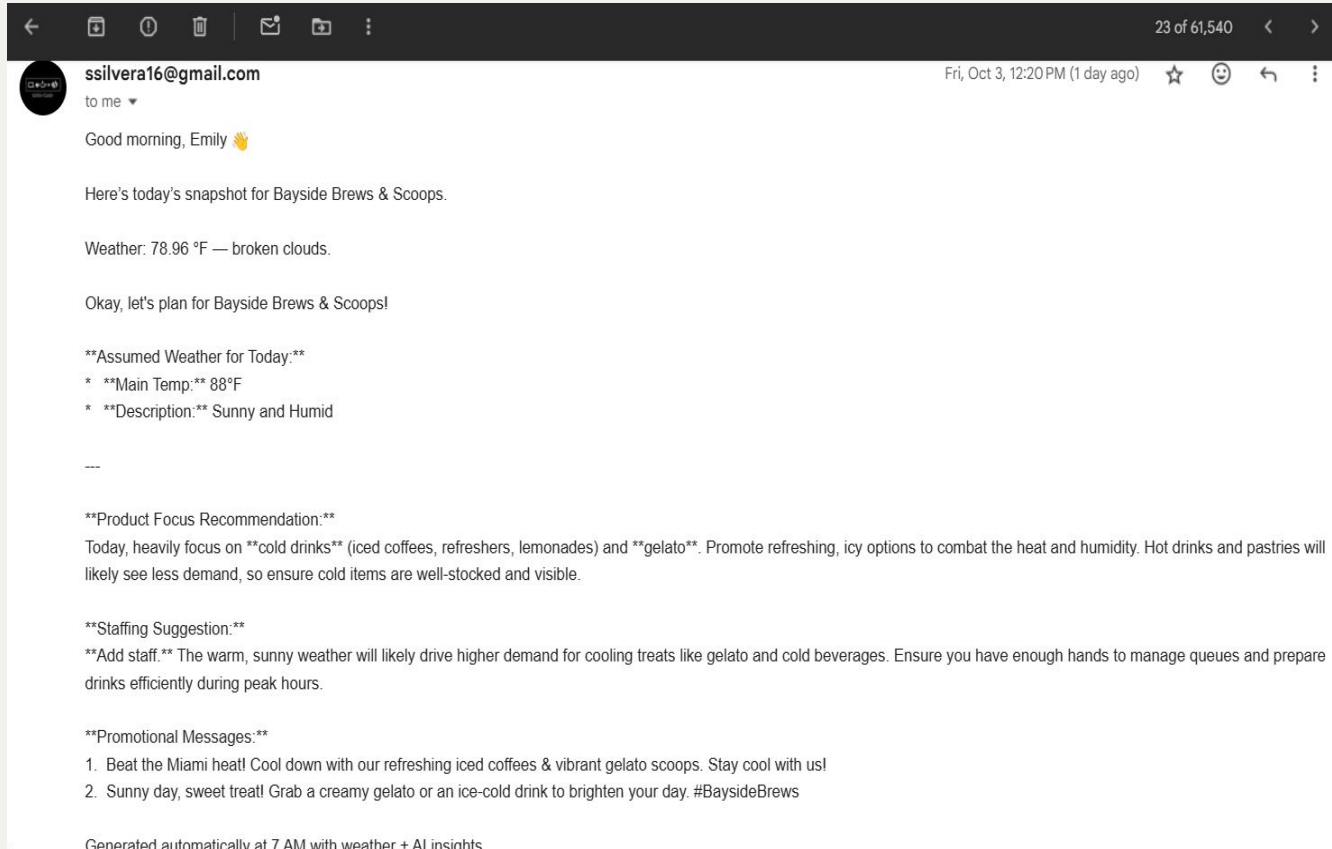
+ Add value set

Send As JSON ⓘ
✓ Yes × No

Continue

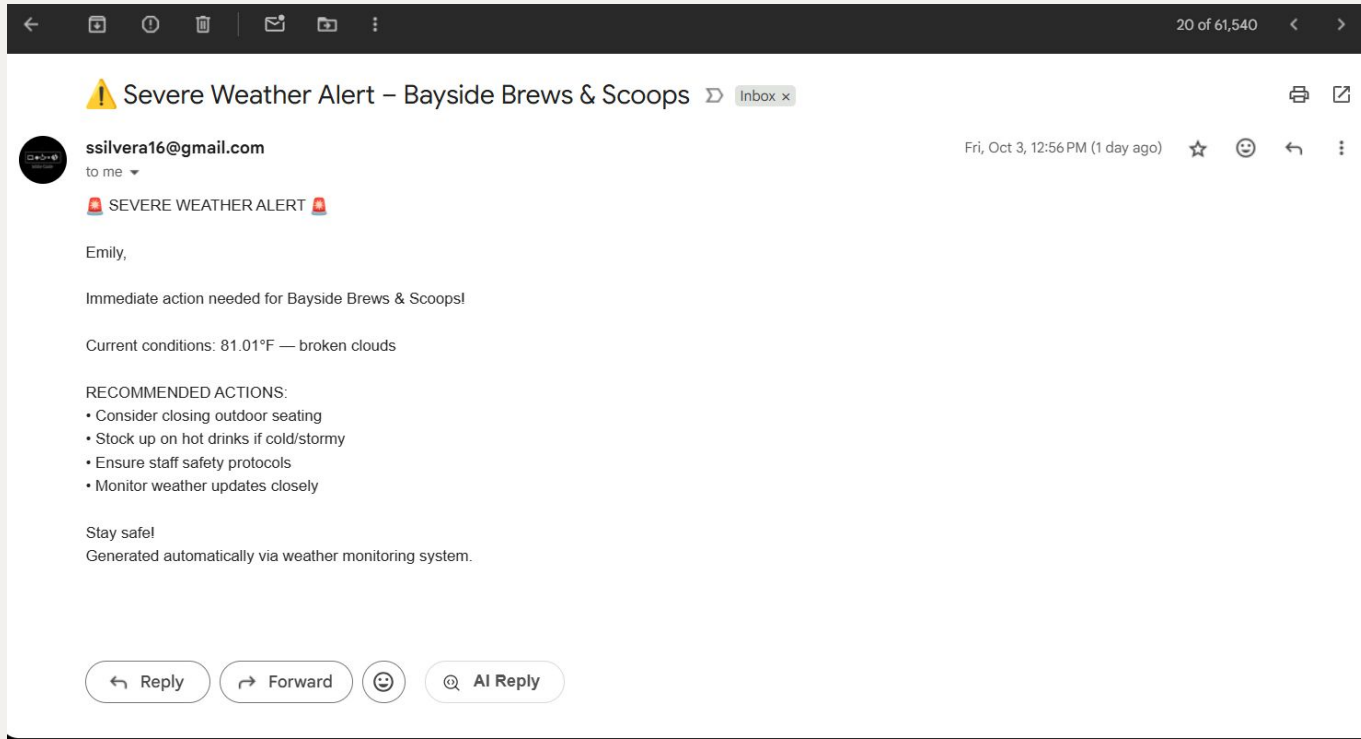
**Miami Kiosk
Weather
Guardian - Zapier
Automation Vid**

Workflow in action (cont.)



Miami Kiosk Weather Guardian - Zapier Automation Vid

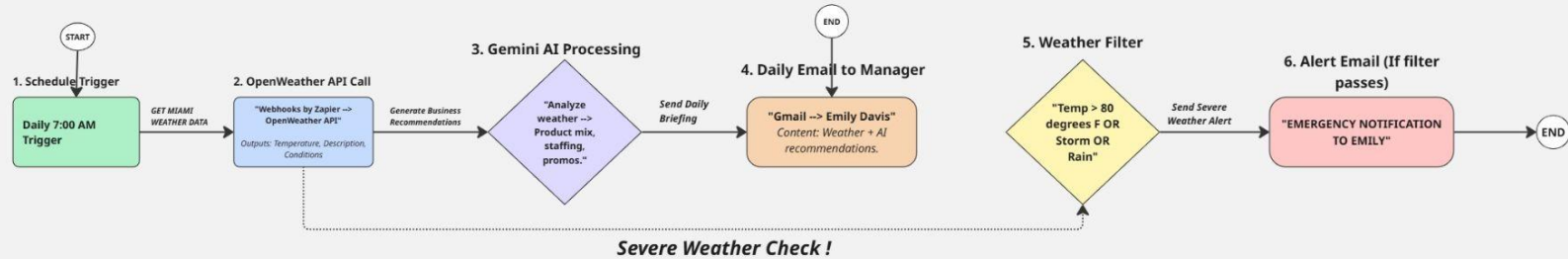
Workflow in action (cont.)



**Miami Kiosk
Weather
Guardian - Zapier
Automation Vid**

Bayside Weather Intelligence System

[System Workflow Link](#)



Benefits

- Time Savings: Eliminate 15-20 minutes of daily manual weather checking, allowing Emily to focus on customer service and operation.
- Faster Decisions: Automated 7:00 AM delivery ensures preparation starts immediately, not after manual checking.
- Consistency: No more missed weather checks due to busy morning or human oversight.

Business Impact

- Increased Revenue: Weather - appropriate product recommendation maximizes sales.
- Better Staffing: AI-driven suggestions to prevent understaffing or overstaffing.
- Customer Experience: Targeted Promotions based on weather conditions to create relevant engaging offers.

Risk Management

- Safety First: Severe weather alerts (>80, storms, rain) ensures quick response to dangerous conditions.
- Proactive Planning: Early morning insights allow time for inventory adjustments and staff scheduling.



Challenges & improvements

First Challenge: Webhooks were not working. I kept modifying the URLs until the data was collected and I could move to the next module. I'm still learning to configuring them more efficiently.

Second Challenge: Once placing the prompt into the gemini node, the email was coming out blank, I received the greeting, but no data from the gemini response. I realized I happened mapped the responses. Once I did that it worked perfectly.

Third Challenge: Was not receiving weather alert email for >91 degrees, so I lowered the temperature to 80, the data was picked up correctly and sent out the alert. (Had to alter / modify temperature to receiving alert)

Conclusions

Project Impact:

This weather automation system transforms Bayside Brews & Scoops from reactive to proactive operations. By combining real-time weather data with AI insights, this solution saves times, increases revenue potential and prioritizes safety.